Homework 5: Monte Carlo methods

Submission

Due: Nov 9th, 2026

Submit the following three files in your directory:

1. `lastname_firstname_hw05.ipynb`: one executed Jupyter notebook containing all code and markdown text;
2. `lastname_firstname_hw05.pdf`: a PDF compiled or exported from the executed notebook; and
3. `requirements.txt`: a short list of the packages needed to run the notebook, with version numbers.

Your notebook must run from top to bottom in a fresh session without manual edits. Use the supplied random seeds. Report requested numerical results with at least three significant digits. Keep written explanations concise, but explain the numerical behavior rather than reporting outputs alone.

Question 1: Random-variable generation and diagnostics (50 points)

In this question, use $N = 20{,}000$ unless stated otherwise. Do not call a prewritten random-number generator for the target distribution when a generation method is specified.

1. **Inverse-transform sampling.** Generate N observations from an exponential distribution with rate $\lambda = 2$ using

$$X = -\frac{\log(1-U)}{\lambda}, \qquad U \sim \text{Uniform}(0,1),$$

and seed 123. Report the empirical mean, variance, and probability $\widehat{\Pr}(X > 2)$. Compare them with

$$\mathbb{E}(X) = \frac{1}{2}, \qquad \text{Var}(X) = \frac{1}{4}, \qquad \Pr(X > 2) = e^{-4}.$$

2. **Acceptance–rejection sampling.** The target density is

$$f(x) = 6x(1-x), \qquad 0 \le x \le 1,$$

which is the Beta$(2,2)$ density. Use the uniform proposal $g(x) = 1$ on $[0,1]$ and seed 456.

- Show that the smallest valid envelope constant is $M = 3/2$ and state the theoretical acceptance probability.
- Implement acceptance–rejection sampling until N values have been accepted.
- Report the number of proposals, empirical acceptance rate, sample mean, sample variance.

Compare the last two quantities with the theoretical values 1/2, 1/20.

3. **Multivariate normal generation.** Use seed 789, $N = 50{,}000$, and

$$\mu = \begin{bmatrix} 1 \\ -1 \\ 0.5 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 1 & 0.7 & 0.2 \\ 0.7 & 2 & 0.4 \\ 0.2 & 0.4 & 0.5 \end{bmatrix}.$$

Generate $Z_i \sim N_3(0, I)$ and construct $X_i = \mu + LZ_i$, where $LL^\top = \Sigma$. Report the empirical mean vector, empirical covariance matrix, and explain how the Cholesky factor creates their dependence.

Question 2: Monte Carlo integration and numerical error (35 points)

Estimate

$$I = \int_0^1 \frac{1}{1+x^2}\, dx = \frac{\pi}{4}.$$

If $U \sim \text{Uniform}(0,1)$ and $h(u) = 1/(1+u^2)$, then $I = \mathbb{E}\{h(U)\}$.

1. Generate 100,000 uniforms using seed 1357. Using the first

   $$N \in \{100,\ 1{,}000,\ 10{,}000,\ 100{,}000\}$$

   observations, report the Monte Carlo estimate, estimated standard error, 95% Monte Carlo confidence interval, and absolute error for each N in a table. Produce a log-log plot of the estimated standard error against N.

2. Please do the same as 1, using importance sampling with the proposal density

   $$g(x) = \frac{1}{\log(2)(1+x)}, \qquad 0 \le x \le 1.$$

   If $U \sim \text{Uniform}(0,1)$, then a draw from g can be generated by

   $$X = G^{-1}(U) = 2^U - 1.$$

3. Approximate the same integral with the composite trapezoidal rule using N equally spaced evaluation points for each value of N from Part 1. You may use `np.trapezoid`. Report the absolute quadrature errors and compare them with the Monte Carlo errors using a log–log plot.

Grading guide

Criterion	Points
Correct implementation and numerical results	50
Sound analysis and interpretation	25
Clear figures, tables, and concise communication	15
Reproducibility and complete submission	10
Total	**100**

Minor differences caused by software versions and Monte Carlo variability are acceptable. Numerical results without written interpretation will receive limited credit.